

LWANM: Learned Weighted Atomic Norm Minimization with Data-driven Weighting Mechanism

Zhuoli Zhang, Zai Yang, Wenlong Wang, and Zhiqiang Wei

School of Mathematics and Statistics

Xi'an Jiaotong University

Xi'an, China

zhangzhuoli@stu.xjtu.edu.cn; yangzai@xjtu.edu.cn; wang1813857265@stu.xjtu.edu.cn; zhiqiang.wei@xjtu.edu.cn

Abstract—Atomic Norm Minimization (ANM) and its weighted variant (WANM) are widely adopted in spectral compressed sensing for their ability to achieve high-precision frequency recovery within the continuous domain. However, their practical utility is often constrained by the prohibitive computational complexity of large-scale semidefinite programming (SDP). Moreover, a major challenge lies in the sensitivity of WANM to manually designed weighting strategies, as effective weights depend on problem-dependent prior information. In this paper, we propose a data-adaptive framework for efficient spectral compressed sensing, termed Learned Weighted Atomic Norm Minimization (LWANM). This approach leverages matrix factorization and learns efficient weighting schemes directly from measurements. Furthermore, we explicitly embed physical model consistency and low-rank structural priors into the training objective to guide the learning process. Numerical experiments demonstrate that LWANM significantly outperforms existing methods in both computational efficiency and reconstruction accuracy.

Index Terms—Spectral compressed sensing, weighted atomic norm minimization, Hankel-Toeplitz model, deep learning, deep unfolding.

I. INTRODUCTION

Spectral compressed sensing [1] refers to the recovery of a spectrally sparse signal and its frequencies from part of its regularly spaced samples in the presence of noise, which is known as spectral super-resolution [2], or line spectral compressed sensing [3]. It has been widely used in direction-of-arrival (DOA) estimation [4] and wireless communication [5].

While model-based spectral compressed sensing methods are supported by rigorous theoretical foundations, classical spectral estimation approaches face significant limitations. FFT-based methods [6] suffer from restricted resolution. Subspace approaches like MUSIC [7] and ESPRIT [8] require numerous snapshots and precise model order, hindering their performance with incomplete data. Compressed sensing addresses these challenges through on-grid, off-grid, and gridless formulations [4], [9], [10]. On-grid approaches are limited by discretization errors known as grid mismatch [4], whereas off-grid methods employ first-order Taylor expansions to mitigate these errors [11], [12]. Conversely, gridless approaches, such as Atomic Norm Minimization (ANM) [13] and its weighted variants (WANM) [14], [15], recover signals directly in the continuous frequency domain. Although ANM achieves high precision, WANM can further enhance reconstruction by incorporating prior information via weighted atomic norms [14]. Nevertheless, the prohibitive computational overhead of SDP solvers, coupled with the reliance of WANM on pre-existing knowledge of the signal, often precludes real-time application. To improve scalability, recent studies have reformulated gridless methods as non-convex structured matrix optimization problems using matrix factorization [16], [17]; however, these methods frequently suffer from poor iteration efficiency.

This work was supported by the National Natural Science Foundation of China under grants 12526209 and 12371464.

Recently, data-based deep learning techniques have been widely used in spectral compressed sensing and demonstrated their superior computational performance. These approaches address signal recovery problems using DNNs with a relatively small number of layers [18], [19]. Nonetheless, the absence of model information in DNNs limits their interpretability and generalizability. Deep unfolding techniques have been proposed to address this issue by substituting each iteration of the original algorithm with a neural network stage [20]. The design enables effective incorporation of model information into DNNs, which enhances the interpretability and generalizability. On-grid compressed sensing-based networks, including LISTA [21] and LISTA-Toeplitz [22], achieve superior reconstruction performance compared with other on-grid compressed sensing methods with significantly fewer iterations. However, they still suffer from grid mismatch. Off-grid unfolding networks, such as those proposed in [23], outperform conventional off-grid methods but rely on first-order Taylor expansions, which introduces modeling error. Recently, gridless unfolding networks have been explored, such as the Hankel-based DHMF [25] and the ANM-based LANM [24]. While LANM and DHMF demonstrate favorable performance, DHMF possesses distinct limitations as the Hankel model constitutes a relaxation of the signal structure. Although ANM employs an accurate model and WANM has been shown to further improve reconstruction by manually incorporating prior information into the weights, LANM is built upon the standard ANM formulation and thus lacks an efficient weighting mechanism.

In this paper, to address the challenge that the weights for WANM depends on the precise specified knowledge of the signal, we propose an efficient weighted atomic norm framework for spectral compressed sensing, wherein the weights is learned directly from the measurements. Compared to LANM, the data-adaptive weighted atomic is adopted to learn and then exploit the underlying signal prior from data. We first formulate a structured matrix optimization problem based on WANM, incorporating a learnable weight matrix to better exploit inherent signal structures and accelerate reconstruction. To solve this problem, we develop a fast gradient descent algorithm leveraged by structured matrix factorization and FFT-based operations. This algorithm is subsequently implemented through deep unfolding, drawing upon the LANM framework, to yield a Learned Weighted Atomic Norm Minimization (LWANM) architecture. By reducing the required number of iterations, this approach further enhances computational efficiency. Numerical results demonstrate that LWANM achieves superior reconstruction accuracy and computational speed compared to state-of-the-art methods.

Notations: Boldface letters denote vectors and matrices. $\mathbb{C}$ denotes the complex number field. The operators $(\cdot)^T$, $(\cdot)^*$, $(\cdot)^H$, $\|\cdot\|_F$, and $\|\cdot\|_2$ represent the transpose, complex conjugate, conjugate transpose, Frobenius norm, and ℓ_2 norm, respectively. The j -th entry of vector

$\mathbf{x}$ is x_j . For a matrix $\mathbf{X}$, and $\text{tr}(\mathbf{X})$ denotes its trace. The diagonal matrix with vector $\mathbf{x}$ on the diagonal is denoted by $\text{diag}(\mathbf{x})$. The identity operator is denoted by $\mathcal{I}$. $\mathbf{X} \geq \mathbf{0}$ means that $\mathbf{X}$ is Hermitian positive semidefinite. $\mathcal{T}$ denotes the Hermitian Toeplitz operator that maps $\mathbf{t} \in \mathbb{C}^n$ to an $n \times n$ Hermitian Toeplitz matrix $\mathcal{T}\mathbf{t}$ whose (i, j) -th ($i \geq j$) is given by t_{i-j+1} . $\mathcal{H}$ denotes the Hankel operator that maps $\mathbf{x} \in \mathbb{C}^{2n-1}$ to an $n \times n$ Hankel matrix $\mathcal{H}\mathbf{x}$ whose (i, j) -th entry is given by x_{i+j-1} . $\mathcal{F}^H$ denotes the adjoint operator of an operator $\mathcal{F}$.

II. PROBLEM FORMULATION AND RELATED WORKS

A. Problem Formulation

A spectrally sparse signal $\mathbf{x} \in \mathbb{C}^N$ is defined as the superposition of a small number of complex sinusoids, modeled as:

$$x_j = \sum_{k=1}^K s_k e^{i2\pi f_k(j-1)}, \quad j = 1, \dots, N, \quad (1)$$

where N and K denote the number of samples and spectral components, respectively, $f_k \in [0, 1)$ represents the normalized frequency and $s_k \in \mathbb{C}$ is the complex amplitude of the k -th component. In practical scenarios, one typically observes only a subset of the entries of $\mathbf{x}$ corrupted by additive noise. Let $\Omega \subset \{1, \dots, N\}$ denote the set of observed indices with cardinality $|\Omega| = M < N$. The observed signal $\mathbf{y}$ is given by:

$$y = \mathcal{P}_\Omega(x + e), \quad (2)$$

where $e \in \mathbb{C}^N$ represents additive noise and $\mathcal{P}_\Omega$ is the sampling operator that retains entries indexed by Ω and sets others to zero. This paper addresses the recovery of the full signal $\mathbf{x}$ and its constituent frequencies from these partial measurements. For simplicity, we assume N is odd, i.e., $N = 2n - 1$ for some positive integer n ; if N is even, the $(N + 1)$ -th entry is treated as a missing measurement.

B. Learned Atomic Norm Minimization (LANM)

LANM [24] focuses on unfolding network implementation for gridless spectral compressed sensing by solving the following ANM:

$$\min_{\mathbf{x}} \lambda \|\mathbf{x}\|_{\mathcal{A}} + \frac{1}{2} \|\mathcal{P}_{\Omega}(\mathbf{x} - \mathbf{y})\|_2^2, \quad (3)$$

where $\lambda > 0$ is the regularization parameter determined by prior knowledge of noise level. The method reformulates ANM using a Hankel–Toeplitz structured model and solves the resulting non-convex problem via gradient descent based on matrix factorization and FFT-based operations, which only requires an upper bound on the model order. To further improve iteration efficiency, denoising convolutional neural networks (DnCNNs) [26] are incorporated to refine intermediate iterates. The overall network architecture follows the unfolding paradigm, where each stage corresponds to one iteration of the underlying optimization algorithm, as illustrated in Fig. 1.

While LANM achieves comparable reconstruction performance and high computational efficiency through fast gradient descent steps implemented via FFTs and a small number of layers, it lacks an efficient weighting mechanism to further exploit data-driven information, which is beneficial for signal and frequency reconstruction.

III. FAST GRADIENT DESCENT ALGORITHM FOR WANM

In this section, to overcome the limitation mentioned above, we propose a WANM framework to learn the underlying weighting scheme from data to improve and accelerate the signal reconstruction and cast it into an unconstrained problem. Then we develop the corresponding fast non-convex gradient descent algorithm.

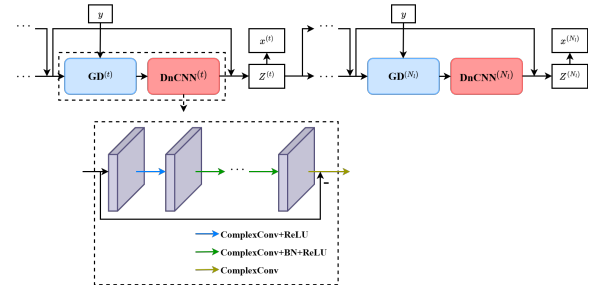


Fig. 1. Structure of LANM [24].

Assuming the weighting function $w_1(f) = \frac{1}{\mathbf{a}(f)^H \mathbf{W}_1 \mathbf{a}(f)}$ for some matrix $\mathbf{W}_1 \geq \mathbf{0}$, the weighted atomic norm is formulated as

$$\|\mathbf{x}\|_{\mathcal{A}^{w_1}} = \inf \left\{ \sum_k \frac{|s_k|}{w_1(f_k)} : \mathbf{x} = \sum_k \mathbf{a}(f_k) s_k \right\}. \quad (4)$$

To incorporate a data-adaptive weighting scheme capable of handling diverse noise profiles, we consider the following WANM problem:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_{\mathcal{A}^{w_1}} + \frac{1}{2} \left\| \mathcal{P}_{\Omega} \frac{\mathbf{W}_2}{\sqrt{\lambda}} \mathcal{D}(\mathbf{x} - \mathbf{y}) \right\|_2^2, \quad (5)$$

where $\mathbf{W}_2 = \text{diag}\{w_{21}, \dots, w_{2N}\}$ with $w_{2i} \geq 0$ denoting unspecified parameters, and $\mathcal{D}\mathbf{x} = [x_1, \sqrt{2}x_2, \dots, \sqrt{n}x_n, \sqrt{n-1}x_{n+1}, \dots, x_N]^T$. To prevent (5) from degenerating, where large eigenvalues of the $\mathbf{W}_1$ would artificially reduce the weighted atomic norm, we constrain the trace of $\mathbf{W}_1$ to be unity.

Low rank Hankel-Toeplitz model provides a new perspective to accurately characterize the spectrally sparse signal [27]. In [28], it has been shown that if $K^* = \text{rank}(\mathcal{T}\mathbf{t}^*) < n$, where $\mathbf{t}^*$ is the optimizer of following SDP:

$$\min_t \text{tr}(\mathbf{W}_1 \mathcal{T} t) \quad \text{s.t.} \quad \begin{bmatrix} \mathcal{T} \bar{t} & \mathcal{H} \bar{x} \\ \mathcal{H} x & \mathcal{T} t \end{bmatrix} \geq 0, \quad (6)$$

then $\|\mathbf{x}\|_{\mathcal{A}^{w_1}} = \text{tr}(\mathbf{W}_1 \mathcal{T} \mathbf{t}^*)$. Inspired by [16], we characterize the weighted atomic norm by prior information of low-rankness in the following theorem:

Theorem 1: For any signal $\mathbf{x}$, let $\mathbf{t}^*$ denote the optimizer of (6), if $K^* = \text{rank}(\mathcal{T}\mathbf{t}^*) \leq R < n$, then for $\mathbf{t} \in \mathbb{C}^n$, $\mathbf{Z} \in \mathbb{C}^{n \times R}$,

$$\|\mathbf{x}\|_{\mathcal{A}^{w_1}} = \min_{\mathbf{t}, \mathbf{Z}} \text{tr}(\mathbf{W}_1 \mathbf{Z} \mathbf{Z}^H) \text{ s.t. } \mathcal{H}\mathbf{x} = \mathbf{Z} \mathbf{Z}^T, \mathcal{T}\mathbf{t} = \mathbf{Z} \mathbf{Z}^H. \quad (7)$$

Proof. Let $\{\mathbf{Z}^*, t^*\}$ denote the optimizer of (7), then we have

$$\begin{bmatrix} \overline{\mathcal{T}t^*} & \mathcal{H}\bar{x} \\ \mathcal{H}x & \overline{\mathcal{T}t^*} \end{bmatrix} = \overline{\begin{bmatrix} Z^* \\ Z^* \end{bmatrix}} \left(\overline{\begin{bmatrix} Z^* \\ Z^* \end{bmatrix}} \right)^H \geq \mathbf{0}, \quad (8)$$

which implies that $\mathbf{t}^*$ is a feasible solution of (6). Thus

$$\|\mathbf{x}\|_{4w_1} \leq \text{tr}(\mathbf{W}_1 \mathcal{T} \mathbf{t}^*) = \text{tr}(\mathbf{W}_1 \mathbf{Z}^* (\mathbf{Z}^*)^H). \quad (9)$$

Next, following the proof of [16, Theorem 1], there exists $\mathbf{Z}^*$, such that

$$\mathcal{H}\mathbf{x} = \mathbf{Z}^*(\mathbf{Z}^*)^T, \mathcal{T}\mathbf{t}^* = \mathbf{Z}^*(\mathbf{Z}^*)^H, \quad (10)$$

which implies $\{\mathbf{Z}^*, \mathbf{t}^*\}$ is a feasible solution of (7). Therefore,

$$\text{tr}(\mathbf{W}_1 \mathbf{Z}^* (\mathbf{Z}^*)^H) \leq \text{tr}(\mathbf{W}_1 \mathcal{T} \mathbf{t}^*) = \|\mathbf{x}\|_{A^{w_1}}, \quad (11)$$

completing the proof. $\square$

Making use of Theorem 1, (5) can be reformulated as

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{t}, \mathbf{Z}} \quad & \text{tr}(\mathbf{W}_1 \mathbf{Z} \mathbf{Z}^H) + \frac{1}{2} \left\| \mathcal{P}_\Omega \frac{\mathbf{W}_2}{\sqrt{\lambda}} \mathcal{D}(\mathbf{x} - \mathbf{y}) \right\|_2^2 \\ \text{s.t.} \quad & \mathcal{H}\mathbf{x} = \mathbf{Z} \mathbf{Z}^T, \mathcal{T}\mathbf{t} = \mathbf{Z} \mathbf{Z}^H. \end{aligned} \quad (12)$$

To facilitate the development of an efficient solution algorithm, we impose an assumption on the range of $\mathbf{W}_1$. Specifically, for any given $\mathbf{W}_1$, one can construct a Toeplitz matrix $\mathbf{W}'_1 = \mathcal{T} \mathcal{D}^{-2} \mathcal{T}^H (\mathbf{W}_1)$ such that $\text{tr}(\mathbf{W}_1 \mathcal{T} \mathbf{t}) = \text{tr}(\mathbf{W}'_1 \mathcal{T} \mathbf{t})$. Moreover, since $\mathbf{W}_1$ is positive semidefinite, we may assume that $\mathbf{W}_1$ is a PSD Toeplitz matrix. Then we consider taking the constraints in (12) as regularization term. Define $\mathcal{G} = \mathcal{H} \mathcal{D}^{-1}$ and $\mathcal{W} = \mathcal{T} \mathcal{D}^{-1}$. Since $\mathcal{G}^H \mathcal{G} = \mathcal{W}^H \mathcal{W} = \mathcal{I}$, $\mathcal{G} \mathcal{G}^H$ and $\mathcal{W} \mathcal{W}^H$ are orthogonal projections onto Hankel and Toeplitz subspaces respectively. Then $\mathbf{Z} \mathbf{Z}^T$ is Hankel and $\mathbf{Z} \mathbf{Z}^H$ is Toeplitz if and only if $(\mathcal{I} - \mathcal{G} \mathcal{G}^H)(\mathbf{Z} \mathbf{Z}^T) = \mathbf{0}$, $(\mathcal{I} - \mathcal{W} \mathcal{W}^H)(\mathbf{Z} \mathbf{Z}^H) = \mathbf{0}$. Therefore the unconstrained optimization problem is

$$\begin{aligned} \min_{\mathbf{Z}} \quad & \frac{1}{p} \text{tr}(\mathbf{W}_1 \mathbf{Z} \mathbf{Z}^H) + \frac{1}{2p} \left\| \mathcal{P}_\Omega \frac{\mathbf{W}_2}{\sqrt{\lambda}} (\mathcal{G}^H(\mathbf{Z} \mathbf{Z}^T) - \mathcal{D}\mathbf{y}) \right\|_2^2 \\ & + \frac{1}{2} \left\| (\mathcal{I} - \mathcal{G} \mathcal{G}^H)(\mathbf{Z} \mathbf{Z}^T) \right\|_F^2 + \frac{1}{2} \left\| (\mathcal{I} - \mathcal{W} \mathcal{W}^H)(\mathbf{Z} \mathbf{Z}^H) \right\|_F^2, \end{aligned} \quad (13)$$

where $p = M/N$ is the sampling ratio. Then we develop a fast gradient descent algorithm to solve (13). Denoting the objective function by $f(\mathbf{Z})$, the Wirtinger gradient and the update step are

$$\begin{aligned} \nabla f(\mathbf{Z}) = & \mathcal{G}((p\lambda)^{-1} \mathcal{P}_\Omega \mathbf{W}_2^2 (\mathcal{G}^H(\mathbf{Z} \mathbf{Z}^T) - \mathcal{D}\mathbf{y}) - \mathcal{G}^H(\mathbf{Z} \mathbf{Z}^T)) \bar{\mathbf{Z}} \\ & - \mathcal{W} \mathcal{W}^H(\mathbf{Z} \mathbf{Z}^H) \mathbf{Z} + \mathbf{Z}(\mathbf{Z}^T \bar{\mathbf{Z}} + \mathbf{Z}^H \mathbf{Z}) + p^{-1} \mathbf{W}_1 \mathbf{Z}. \\ \mathbf{Z}^{(t)} = & \mathbf{Z}^{(t-1)} - \eta^{(t-1)} \nabla f(\mathbf{Z}^{(t-1)}). \end{aligned} \quad (14)$$

Then the estimate $\hat{\mathbf{x}} = \mathcal{D}^{-1} \mathcal{G}^H(\hat{\mathbf{Z}} \hat{\mathbf{Z}}^T)$. Note that $(\mathcal{G}\mathbf{v})\mathbf{Z}, (\mathcal{W}\mathbf{v})\mathbf{Z}, \mathcal{G}^H(\mathbf{Z} \mathbf{Z}^T), \mathcal{W}^H(\mathbf{Z} \mathbf{Z}^H)$ and $\mathbf{W}_1 \mathbf{Z}$ can be computed by convolution and FFTs [29] by $O(RN \log N)$ flops and $\mathbf{Z}(\mathbf{Z}^T \bar{\mathbf{Z}} + \mathbf{Z}^H \mathbf{Z})$ can be computed in $O(R^2 N)$ flops. The computational complexity of $\nabla f(\mathbf{Z})$ is $O(RN \log N + R^2 N)$. The initialization of the algorithm is obtained by taking the partial Takagi factorization [30] of the rank- R approximation of $p^{-1} \mathcal{H}\mathbf{y}$, namely $\mathbf{U} \mathbf{\Sigma} \mathbf{U}^T = \Gamma_R(p^{-1} \mathcal{H}\mathbf{y})$, $\mathbf{Z}_0 = \mathbf{U} \mathbf{\Sigma}^{1/2}$, where Γ_R denotes the hard threshold operator. It is worth noting that the computational complexity of Γ_R for Hankel matrices can be reduced to $O(RN \log N)$ [31]; consequently, the proposed algorithm is computationally efficient.

IV. ALGORITHM REALIZATION AND TRAINING STRATEGY

In this section, we present an efficient realization and training strategy for the proposed data-adaptive weighted atomic norm algorithm. The realization is based on an unfolding implementation that follows the same stage-wise structure as LANM [24], where each stage corresponds to one iteration of the gradient-based solver derived in the previous section. The key difference lies in the learnable weighting formulation and the model-consistent training objective.

To ensure the positive semidefinite and Toeplitz structure of the learnable weighting matrix while maintaining computational efficiency, we adopt the following reparameterization in an approximate sense based on the Vandermonde decomposition of PSD Toeplitz matrices [32], [33].

Lemma 1: For any PSD Toeplitz matrix $\mathcal{T}\mathbf{t} \in \mathbb{C}^{n \times n}$, there exists a sequence $\{x_l\}$, such that $\mathcal{T}\mathbf{t}_L \rightarrow \mathcal{T}\mathbf{t}$, $L \rightarrow \infty$, where the elements

of $\mathbf{t}_L$ are defined as $t_{L,j} = \frac{1}{L} \sum_{l=1}^{L-j+1} \bar{x}_l x_{l+j-1}$ and the matrix $\mathcal{T}\mathbf{t}_L$ can be factored as:

$$\mathcal{T}\mathbf{t}_L = \mathbf{M}_L \mathbf{M}_L^H, \quad \mathbf{M}_L = \begin{bmatrix} \frac{x_1}{\sqrt{L}} & \cdots & \frac{x_L}{\sqrt{L}} & & & \\ & \frac{x_1}{\sqrt{L}} & \cdots & \frac{x_L}{\sqrt{L}} & & \\ & & \ddots & \vdots & \ddots & \\ & & & \frac{x_1}{\sqrt{L}} & \cdots & \frac{x_L}{\sqrt{L}} \end{bmatrix}, \quad (15)$$

where $\mathbf{M}_L \in \mathbb{C}^{n \times (L-n+1)}$ is a rectangular Toeplitz matrix.

Proof. According to the Vandermonde decomposition of $\mathcal{T}\mathbf{t}$, the entries are given by $t_j = \sum_{k=1}^K e^{i2\pi f_k p_k} p_k$, $j = 1, \dots, n$ for some $p_k > 0$, $K = \text{rank}(\mathcal{T}\mathbf{t}) \leq n$. Let $x_l = \sum_{k=1}^K e^{i2\pi l f_k} \sqrt{p_k}$ and $t_{L,j} = \frac{1}{L} \sum_{l=1}^{L-j+1} \bar{x}_l x_{l+j-1}$. Substituting x_l into $t_{L,j}$, we have

$$\begin{aligned} t_{L,j} = & \left(1 - \frac{j-1}{L}\right) t_j \\ & + \frac{1}{L} \sum_{k_1 \neq k_2} \sqrt{p_{k_1} p_{k_2}} e^{i2\pi(j-1)f_{k_2}} \sum_{l=1}^{L-j+1} e^{i2\pi l(f_{k_2} - f_{k_1})} \end{aligned} \quad (16)$$

For a fixed set of frequencies $\{f_k\}_{k=1}^K$, the second term in (16) vanishes as $L \rightarrow \infty$; thus, $t_{L,j} \rightarrow t_j$ and $\mathcal{T}\mathbf{t}_L \rightarrow \mathcal{T}\mathbf{t}$. Defining $\mathbf{M}_L$ as in (15), we obtain the factorization $\mathcal{T}\mathbf{t}_L = \mathbf{M}_L \mathbf{M}_L^H$. $\square$

Making use of Lemma 1, PSD Toeplitz matrix $\mathbf{W}_1$ in (12) can be approximated by $\mathbf{M}_L \mathbf{M}_L^H$ given by (15). Note that $\mathbf{M}_L$ also admits Toeplitz structure, enabling fast computation of $\mathbf{M}_L \mathbf{M}_L^H \mathbf{Z}$ in $O(LR \log L)$ flops. In the unfolding implementation, $\mathbf{M}_L$ and $\mathbf{W}_2$ are treated as learnable parameters and are stage-specific, allowing each stage to have its own independent set of parameters.

To further reduce the number of iterations required by the gradient descent algorithm, we adopt the unfolding-based implementation following LANM [24]. Each unfolding stage corresponds to one iteration of the proposed solver, where a DnCNN module is employed to refine the gradient update and a learnable momentum term is used to accelerate convergence:

$$\begin{aligned} \mathbf{V}^{(k+1)} = & \text{DnCNN}^{(k)} \left(\mathbf{Z}^{(k)} - \eta^{(k)} \nabla f(\mathbf{Z}^{(k)}) \right), \\ \mathbf{Z}^{(k+1)} = & \alpha^{(k)} \mathbf{V}^{(k+1)} + \beta^{(k)} \mathbf{V}^{(k)}, \end{aligned} \quad (17)$$

where $\alpha^{(k)}$ and $\beta^{(k)}$ are learnable parameters and a learnable piecewise linear function is employed to model the underlying relationship between the regularization parameter and the noise level. The overall unfolding structure is identical to that of LANM illustrated in Fig. 1.

Compared to LANM, we explicitly incorporate signal structure and forward-model consistency into the loss function to increase generalizability and robustness to noise. The resulting loss function is given by:

$$\begin{aligned} L(\Theta) = & \frac{1}{N_t} \frac{1}{N_l} \sum_{i=1}^{N_t} \sum_{j=1}^{N_l} \left[\left\| \hat{\mathbf{x}}_i^{(j)}(\Theta) - \mathbf{x}_i^{\text{gt}} \right\|_2^2 \right. \\ & + \alpha \text{tr}(\hat{\mathbf{Z}}_i^{(j)}(\Theta) \hat{\mathbf{Z}}_i^{(j)}(\Theta)^H) \\ & + \beta \left(\left\| (\mathcal{I} - \mathcal{G} \mathcal{G}^H)(\hat{\mathbf{Z}}_i^{(j)}(\Theta) \hat{\mathbf{Z}}_i^{(j)}(\Theta)^T) \right\|_F^2 \right. \\ & \left. + \left\| (\mathcal{I} - \mathcal{W} \mathcal{W}^H)(\hat{\mathbf{Z}}_i^{(j)}(\Theta) \hat{\mathbf{Z}}_i^{(j)}(\Theta)^H) \right\|_F^2 \right) \\ & \left. + \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{\gamma}{\sigma_i^2} \left\| \mathcal{P}_\Omega(\hat{\mathbf{x}}_i^{(j)}(\Theta) - \mathbf{y}_i) \right\|_2^2, \right] \end{aligned} \quad (18)$$

where N_t and N_l denote the number of training samples and unfolding stages, respectively, $\mathbf{x}_i^{\text{gt}}$ is the ground-truth signal, and σ_i^2 denotes the noise variance of the i -th training sample. The terms in

the loss function correspond to signal fidelity, sparsity promotion, Hankel–Toeplitz structural consistency, and forward measurement consistency, respectively. By minimizing this loss, the unfolding network is trained in a manner guided by the underlying signal model and structural priors.

V. SIMULATION RESULTS

In this section, we conduct an ablation study on the network structure and loss function, and evaluate the reconstruction performance and computational efficiency of the proposed network compared to ANM-ADMM [13], ANM-HT-GD [24], LISTA-Toeplitz [22] and LANM [24]. The maximum number of iterations of ANM-ADMM and ANM-HT-GD is set to 10^3 , and the stopping criterion for ANM-HT-GD is $\|\hat{\mathbf{x}}^{(t)} - \hat{\mathbf{x}}^{(t-1)}\|_2 / \|\hat{\mathbf{x}}^{(t-1)}\|_2 < 10^{-4}$. The number of stages for LISTA-Toeplitz, LANM, and LWANM is set to 10. The size of the preset grid of LISTA-Toeplitz is set to 512. The training set for all three networks consists of 3.5×10^5 samples with $K = 1, 2, 3, 4, 5$, SNR = 0, 5, 10, 15, 20, 25, 30dB and sampling ratio of 0.2. The rank parameter R of ANM-HT-GD, LANM and the proposed network is set to 30 as conservative upper bound for practical robustness since in real-world scenarios the true model order is often difficult to estimate reliably from incomplete and noisy observations.

In *Experiment 1*, we examine the efficiency of the network structure and the loss function. LWANM-Enh denotes the proposed network with the physics-informed loss in (18). LWANM refers to the same as LWANM-Enh, but using the standard loss function in [24], i.e., $\alpha = \beta = \gamma = 0$. DeepCNN denotes the network illustrated in Fig. 1 with the GD modules removed and uses the same loss function as LWANM-Enh. The parameters are set to $N = 255$, $M = 0.2N$, $K = 5$ and the frequencies are randomly spaced with minimum separation of $\Delta = 1.5/N$ in ascending order and f_2 is fixed at $f_1 + \Delta$. The simulation is conducted over 20 Monte Carlo trials. Fig. 2 illustrates the frequency recovery and signal reconstruction performance. The results indicate that both the model-based GD module in LWANM-Enh and the physics-informed loss function facilitate signal and frequency recovery. For simplicity, we refer to LWANM-Enh as LWANM in the sequel.

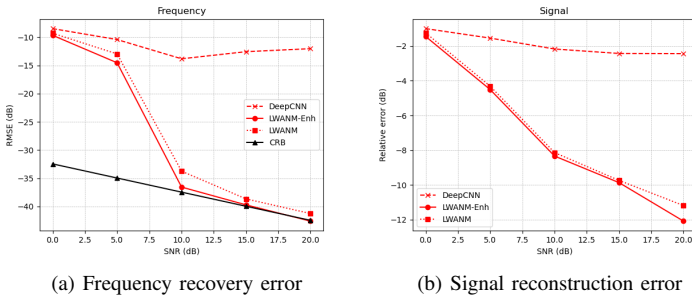


Fig. 2. Ablation study on (a) frequency recovery and (b) signal reconstruction.

In *Experiment 2*, we evaluate the robustness to noise across different noise levels using 20 Monte Carlo trials. Signal parameters are set to $N = 255$, $M = 0.2N$, $K = 5$. The frequencies f_i are arranged in ascending order with a minimum separation of $\Delta = 1.5/N$, and f_2 is fixed at $f_1 + \Delta$. As shown in Fig. 3(a) LISTA-Toeplitz exhibits a higher MSE than gridless methods in high SNR due to the grid mismatch issue and low iteration efficiency of ISTA. At high SNRs, LANM and ANM-HT-GD do not outperform ANM-ADMM, because the relatively loose rank parameter limits the effectiveness

of low-rank promotion in each stage and iteration. In contrast, ANM-ADMM achieves the best performance while LWANM exhibits slightly inferior yet comparable performance. Fig. 3(b) demonstrates the signal reconstruction performance of different algorithms. Owing to the sparsity-promoting atomic norm and weighted atomic norm regularization, model-based methods tend to favor solutions with smaller signal amplitudes to reduce the objective value, which leads to inferior signal reconstruction accuracy while preserving comparable frequency estimates. In contrast, LWANM, LANM, and LISTA-Toeplitz alleviate this bias through direct optimization of the signal-domain training objective.

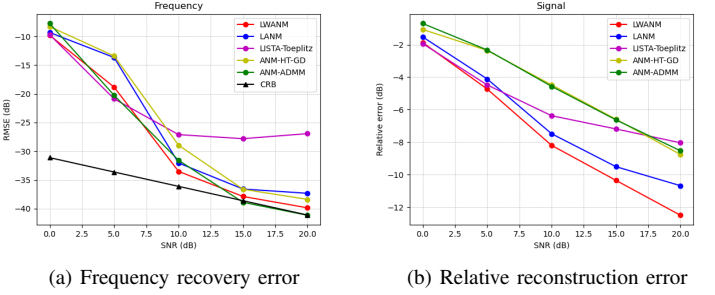


Fig. 3. (a) Root mean-squared error of frequencies and (b) relative reconstruction error of the signal across different SNRs.

In *Experiment 3*, we compare the computational efficiency by proportionally increasing problem size using 10 Monte Carlo trials when $M = 0.2N$, $K = 5$. We present the result of computational efficiency at 10dB in Fig. 4. ANM-ADMM is the most time-consuming due to the eigen-decomposition in each iteration. ANM-HT-GD takes more iterations to converge though it has low iteration complexity. Partly due to the introduction of DnCNN structure, the computational time of LWANM and LANM is larger than LISTA-Toeplitz which does not introduce any neural network structure.

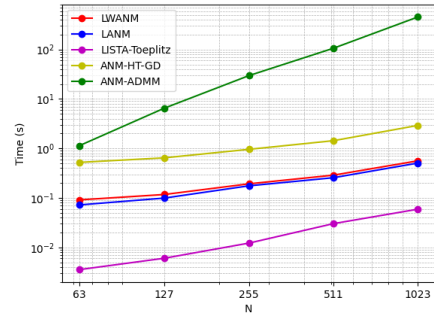


Fig. 4. Computational time under different N .

VI. CONCLUSION

In this paper, we first proposed an efficient data-adaptive WANM framework based on Hankel–Toeplitz matrix factorization. We then realize an unfolding implementation following LANM referred to as LWANM to further improve computational efficiency. Numerical simulations demonstrated that LWANM achieved comparable reconstruction performance compared to other methods while significantly reducing computational time.

REFERENCES

- [1] M. F. Duarte and R. G. Baraniuk, "Spectral compressive sensing," *Applied and Computational Harmonic Analysis*, vol. 35, no. 1, pp. 111–129, 2013.
- [2] E. J. Candès and C. Fernandez-Granda, "Towards a mathematical theory of super-resolution," *Communications on Pure and Applied Mathematics*, vol. 67, no. 6, pp. 906–956, 2014.
- [3] Z. Yang and L. Xie, "On gridless sparse methods for line spectral estimation from complete and incomplete data," *IEEE Transactions on Signal Processing*, vol. 63, no. 12, pp. 3139–3153, 2015.
- [4] Z. Yang, J. Li, P. Stoica, and L. Xie, "Sparse methods for direction-of-arrival estimation," in *Academic Press Library in Signal Processing, Volume 7*. Elsevier, 2018, pp. 509–581.
- [5] Y. Tsai, L. Zheng, and X. Wang, "Millimeter-wave beamformed full-dimensional mimo channel estimation based on atomic norm minimization," *IEEE Transactions on Communications*, vol. 66, no. 12, pp. 6150–6163, 2018.
- [6] E. O. Brigham, *The fast Fourier transform and its applications*. Prentice-Hall, Inc., 1988.
- [7] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, vol. 34, no. 3, pp. 276–280, 1986.
- [8] R. Roy and T. Kailath, "Esprit-estimation of signal parameters via rotational invariance techniques," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 7, pp. 984–995, 2002.
- [9] E. J. Candès, J. Romberg, and T. Tao, "Robust uncertainty principles: Exact signal reconstruction from highly incomplete frequency information," *IEEE Transactions on Information Theory*, vol. 52, no. 2, pp. 489–509, 2006.
- [10] M. F. Duarte and Y. C. Eldar, "Structured compressed sensing: From theory to applications," *IEEE Transactions on Signal Processing*, vol. 59, no. 9, pp. 4053–4085, 2011.
- [11] Z. Yang, C. Zhang, and L. Xie, "Robustly stable signal recovery in compressed sensing with structured matrix perturbation," *IEEE Transactions on Signal Processing*, vol. 60, no. 9, pp. 4658–4671, 2012.
- [12] Z. Yang, L. Xie, and C. Zhang, "Off-grid direction of arrival estimation using sparse bayesian inference," *IEEE Transactions on Signal Processing*, vol. 61, no. 1, pp. 38–43, 2013.
- [13] B. N. Bhaskar, G. Tang, and B. Recht, "Atomic norm denoising with applications to line spectral estimation," *IEEE Transactions on Signal Processing*, vol. 61, no. 23, pp. 5987–5999, 2013.
- [14] Z. Yang and L. Xie, "Fast convex optimization method for frequency estimation with prior knowledge in all dimensions," *Signal Processing*, vol. 142, pp. 271–280, 2018.
- [15] K. V. Mishra, M. Cho, A. Kruger, and W. Xu, "Spectral super-resolution with prior knowledge," *IEEE Transactions on Signal Processing*, vol. 63, no. 20, pp. 5342–5357, 2015.
- [16] X. Wu, Z. Yang, J.-F. Cai, and Z. Xu, "Spectral super-resolution on the unit circle via gradient descent," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [17] H. Cai, J.-F. Cai, and J. You, "Structured gradient descent for fast robust low-rank hankel matrix completion," *SIAM Journal on Scientific Computing*, vol. 45, no. 3, pp. A1172–A1198, 2023.
- [18] Z.-M. Liu, C. Zhang, and S. Y. Philip, "Direction-of-arrival estimation based on deep neural networks with robustness to array imperfections," *IEEE Transactions on Antennas and Propagation*, vol. 66, no. 12, pp. 7315–7327, 2018.
- [19] X. Wu, X. Yang, X. Jia, and F. Tian, "A gridless doa estimation method based on convolutional neural network with toeplitz prior," *IEEE Signal Processing Letters*, vol. 29, pp. 1247–1251, 2022.
- [20] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Processing Magazine*, vol. 38, no. 2, pp. 18–44, 2021.
- [21] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proceedings of the 27th International Conference on Machine Learning*, 2010, pp. 399–406.
- [22] R. Fu, Y. Liu, T. Huang, and Y. C. Eldar, "Structured lista for multidimensional harmonic retrieval," *IEEE Transactions on Signal Processing*, vol. 69, pp. 3459–3472, 2021.
- [23] X. Su, Z. Liu, J. Shi, P. Hu, T. Liu, and X. Li, "Real-valued deep unfolded networks for off-grid doa estimation via nested array," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 4, pp. 4049–4062, 2023.
- [24] Z. Yang, Z. Zhang, W. Wang, W. Zheng, Y. Yang, and Z. Wei, "Lanm: Learned atomic norm minimization for superfast gridless spectral compressed sensing," in *2025 IEEE 35th International Workshop on Machine Learning for Signal Processing (MLSP)*. IEEE, 2025, pp. 1–6.
- [25] Y. Huang, J. Zhao, Z. Wang, V. Orehov, D. Guo, and X. Qu, "Exponential signal reconstruction with deep hankel matrix factorization," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 9, pp. 6214–6226, 2021.
- [26] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising," *IEEE Transactions on Image Processing*, vol. 26, no. 7, pp. 3142–3155, 2017.
- [27] X. Wu, Z. Yang, P. Stoica, and Z. Xu, "Maximum likelihood line spectral estimation in the signal domain: A rank-constrained structured matrix recovery approach," *IEEE Transactions on Signal Processing*, vol. 70, pp. 4156–4169, 2022.
- [28] Y. Chu, Z. Wei, and Z. Yang, "New reweighted atomic norm minimization approach for line spectral estimation," *Signal Processing*, vol. 206, p. 108897, 2023.
- [29] J.-F. Cai, T. Wang, and K. Wei, "Fast and provable algorithms for spectrally sparse signal reconstruction via low-rank hankel matrix completion," *Applied and Computational Harmonic Analysis*, vol. 46, no. 1, pp. 94–121, 2019.
- [30] A. M. Chebotarev and A. E. Teretenkov, "Singular value decomposition for the takagi factorization of symmetric matrices," *Applied Mathematics and Computation*, vol. 234, pp. 380–384, 2014.
- [31] D. Potts and M. Tasche, "Fast esprit algorithms based on partial singular value decompositions," *Applied Numerical Mathematics*, vol. 88, pp. 31–45, 2015.
- [32] C. Carathéodory and L. Fejér, "Über den zusammenhang der extremen von harmonischen funktionen mit ihren koeffizienten und über den picard-landau'schen satz," *Rendiconti del Circolo Matematico di Palermo (1884-1940)*, vol. 32, no. 1, pp. 218–239, 1911.
- [33] Z. Yang, L. Xie, and P. Stoica, "Vandermonde decomposition of multilevel toeplitz matrices with application to multidimensional super-resolution," *IEEE Transactions on Information Theory*, vol. 62, no. 6, pp. 3685–3701, 2016.